

AARUSH MODI

amodi32@uwo.ca | linkedin.com/in/aarush-modi | github.com/aarush-modi | aarushmodi.ca

EDUCATION

University of Western Ontario

London, ON

BESc, Software Engineering (Co-op) — Specialization in Artificial Intelligence

Sept. 2024 – June 2029

– Coursework (2026–27): Introduction to Machine Learning (DS 3000); Data Engineering & Machine Learning (AISE 3010)

PROJECTS

Autonomous Investment Committee | *Python, Anthropic Claude API, asyncio, ChromaDB* Mar. 2026 – Aug. 2026

- Designed, built, and shipped solo a multi-agent AI system where 6 specialized LLM agents collaborate as an investment committee: one CLI command takes a ticker and a hypothesis, and returns a structured, citation-backed investment memo with a Buy/Sell/Hold recommendation
- Engineered LLM-driven dynamic orchestration: a Claude “Chair” agent decides at runtime which of 5 specialists (market data, research/RAG, alternative data, quant, risk) to consult via a `delegate_to_agent` tool so the execution graph is chosen by model reasoning, not a hard-coded pipeline
- Built the agent framework from scratch on the raw Anthropic SDK (no LangChain): an async tool-use loop base class shared by all agents, plus a `@tool` decorator generating tool JSON schemas from Python type hints (~15 tools, ~40 modules)
- Parallelized execution with `asyncio`, running up to 5 sub-agents concurrently; added rate-limit backoff honoring `retry-after` and graceful degradation so one agent’s failure never cancels its siblings
- Cut input-token cost ~90% on later loop iterations via prompt caching (rolling `cache_control` breakpoints) and a tiered model architecture — a high-reasoning model for synthesis, cheap fast models for specialists ensuring a full committee run under \$1 in API spend
- Architected the RAG pipeline: structure-aware chunking splits SEC filings along their section hierarchy into a ChromaDB vector store (metadata-filtered retrieval, local sentence-transformer embeddings) so retrieved chunks align with document semantics rather than fixed-size windows
- Decoupled the data layer behind abstract provider interfaces (adapter pattern) with swappable implementations (Yahoo Finance, FRED, SEC EDGAR) so new data sources integrate without touching agent code, backed by an SQLite response cache with per-data-type TTLs

Momentum Backtesting Engine | *Python, pandas, yfinance, SQLite, Matplotlib*

May 2025 – Aug. 2025

- Built a modular backtesting framework for long/short momentum strategies across 1,000+ equities and 20 years of data (5,000,000+ price records), reporting annualized return, Sharpe, volatility, and max drawdown with position sizing, transaction costs, and execution lag modeled

ML Stock Portfolio Analyzer | *Python, pandas, scikit-learn, yfinance, SQLite*

May 2025 – Aug. 2025

- Automated equity data acquisition, storage, and short-term price forecasting for 500+ tickers (100,000+ data points); trained scikit-learn regression models that beat moving-average baselines by 15–20% on MAE/RMSE with sub-second inference

EXPERIENCE

A&A Optoelectronics

Toronto, ON

Software Developer (designated developer)

May 2026 – August 2026

- Sole software developer for the company, leading requirements and design of a new corporate website and a commercial lighting-control app which automates multiple light fixtures over Bluetooth mesh with sensor-driven behaviours like motion-triggered dim timers
- Solely responsible for the live production site’s security and availability: hardening the site, reviewing traffic and access logs, and acting as the company’s single point of contact for technical issues
- Resolved a production outage affecting VPN and secure-browser clients within 24 hours: traced an antivirus malware flag to an unauthorized favicon file in the database, uncovered 1,442 data-access attempts from a single IP, blacklisted the IP, purged the file, and filed the antivirus false-positive report

Career Education Council — program in partnership with Apple

Guelph, ON

iOS App Developer (Co-op)

Oct. 2023 – Jan. 2024

- Selected for an Apple-partnered co-op program in optimized, production-ready iOS app development and entrepreneurship
- Independently researched, designed, and built a self-improvement iOS app in Swift; pitched it at the Guelph Developer Showcase, generating 6 investor inquiries

Privately Managed Residential Properties

Ajax, ON

Property Manager

June 2023 – Oct. 2025

- Independently ran operations, RTA-compliant maintenance, leasing tours, and tenant relations for multiple residential properties on behalf of a non-resident owner for 2+ years, alongside full-time studies

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, Swift, C/C++, MATLAB

AI/LLM Engineering: Anthropic Claude API (tool use), multi-agent orchestration, RAG, embeddings (sentence-transformers), vector databases (ChromaDB), prompt caching, MCP servers, structured outputs (Pydantic), scikit-learn

Frameworks & Libraries: FastAPI, React, Node.js, asyncio, pandas, NumPy, SciPy, Matplotlib, Jinja2

Developer Tools: Git, Docker, VS Code, IntelliJ, Kubernetes, Claude Code